

Assignment - 2

- ① Two Stack PDA for
 $x_2 = \{a^n b^n c^n \mid n \geq 1\}$

Stack 1 (CS1) : Stores a's

Stack 2 (CS2) : Stores b's for matching with c

Transition representation :

$\delta(q_0, a, z_1, z_2) \rightarrow (q_0, q_2, 1, z_2)$

$\delta(q_0, b, a, z_2) \rightarrow (q_1, \epsilon, b z_2)$

$\delta(q_1, b, a, z_2) \rightarrow (q_1, \epsilon, b z_2)$

$\delta(q_1, c, z_1, b) \rightarrow (q_2, z_1, \epsilon)$

$\delta(q_2, c, z_1, b) \rightarrow (q_2, z_1, \epsilon)$

$\delta(q_2, \epsilon, z_1, z_2) \rightarrow (q_f, z_1, z_2)$

Input	Stack 1	Stack 2
a	a	-
a	aa	-
b	a	b
b	-	bb
c	-	b
c	-	-

④

Expression :

$$a = (b + c) * (d - e) + f$$

①

$$t_1 = b + c$$

$$t_2 = d - e$$

$$t_3 = t_1 * t_2$$

$$t_4 = t_3 + f$$

$$a = t_4$$

②

op	Arg 1	Arg 2	Result
+	b	c	t_1
-	d	e	t_2
*	t_1	t_2	t_3
+	t_3	f	t_4
=	t_4	-	a

Symbol table

name	Type	category
sum	int	variable
a	int	variable
b	int	variable

flow :-

int $\rightarrow$ KEYWORD
sum $\rightarrow$ IDENTIFIER
= $\rightarrow$ operator
a $\rightarrow$ IDENTIFIER
+ $\rightarrow$ OPERATOR
b $\rightarrow$ IDENTIFIER
* $\rightarrow$ OPERATOR
10 $\rightarrow$ CONSTANT
; $\rightarrow$ SEPARATOR

output :-

<KEYWORD, int>
<ID, sum>
<ASSIGN_OP, =>
<ID, a>
<PLUS, +>
<ID, b>
<MUL, *>
<NUM, 10>
<SEMI_COLON, ;>

③ Lexical Analysis in a Compiler

* working exploration

- ① Takes source code as input
- ② Removes white spaces & comments
- ③ Breaks program into tokens
- ④ stores identifiers in Symbol Table
- ⑤ sends tokens to syntax analyzer

Example

int sum = a + b * 10;

① Tokens, lexemes, Patterns

Token Type	Lexeme	Pattern
Keyword	int	reserved word.
Identifier	sum	letter
operator	=	assignment
Identifier	a	letter
operator	+	arithmetic
Identifier	b	letter
operator	*	arithmetic
constant	10	digit +
separator	;	symbol.

(d) I_4

$L \rightarrow id \cdot$

(e) I_5

$S \rightarrow L \cdot R$

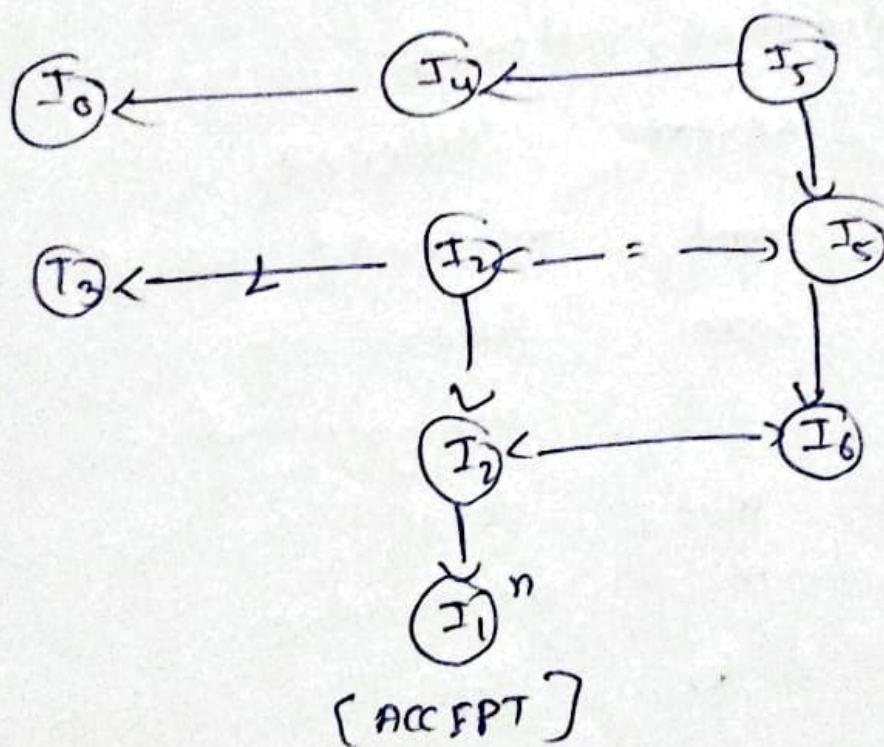
$R \rightarrow \cdot L$

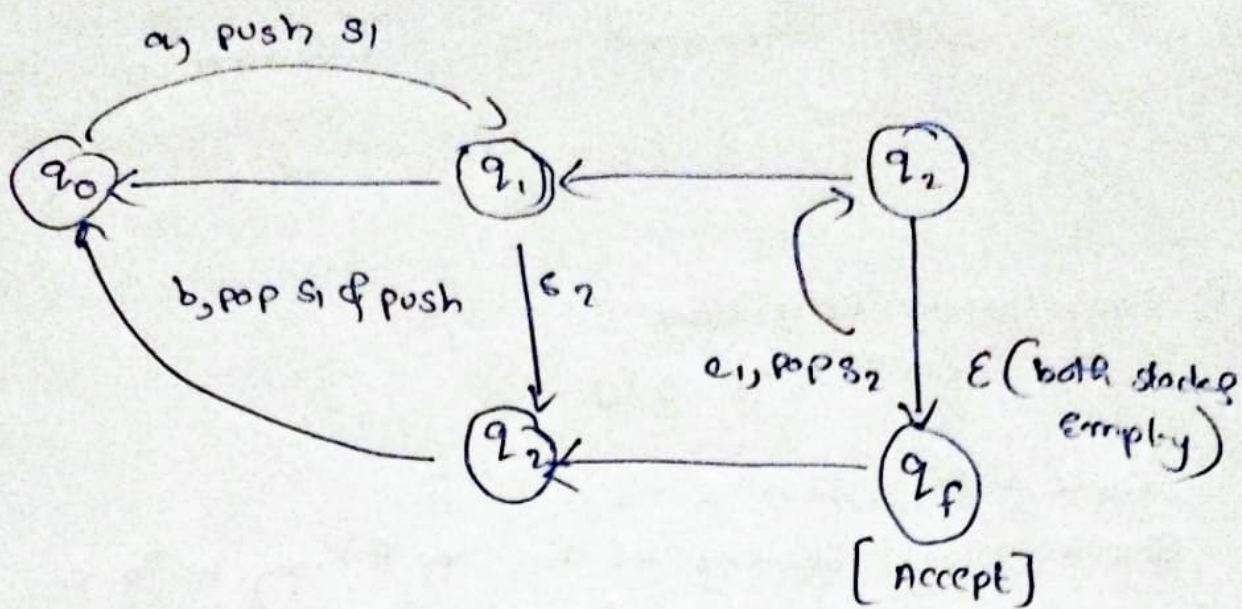
$L \rightarrow \cdot * L$

$L \rightarrow \cdot id$

(f) I_6

$S \rightarrow L = R \cdot$





②

$$S \rightarrow L = R$$

$$L \rightarrow^* L = |id$$

$$R \rightarrow L$$

Augmented Grammar
 $S' \rightarrow S$

① I_0

$$S' \rightarrow \cdot S$$

$$S \rightarrow \cdot L = R$$

$$L \rightarrow \cdot^* L$$

$$L \rightarrow \cdot id$$

② I_1

$$S' \rightarrow S \cdot$$

③ I_3

$$L \rightarrow * \cdot L$$

$$L \rightarrow \cdot * L$$

$$L \rightarrow \cdot id$$